

TECHNICAL DOCUMENTATION — CITABILITY

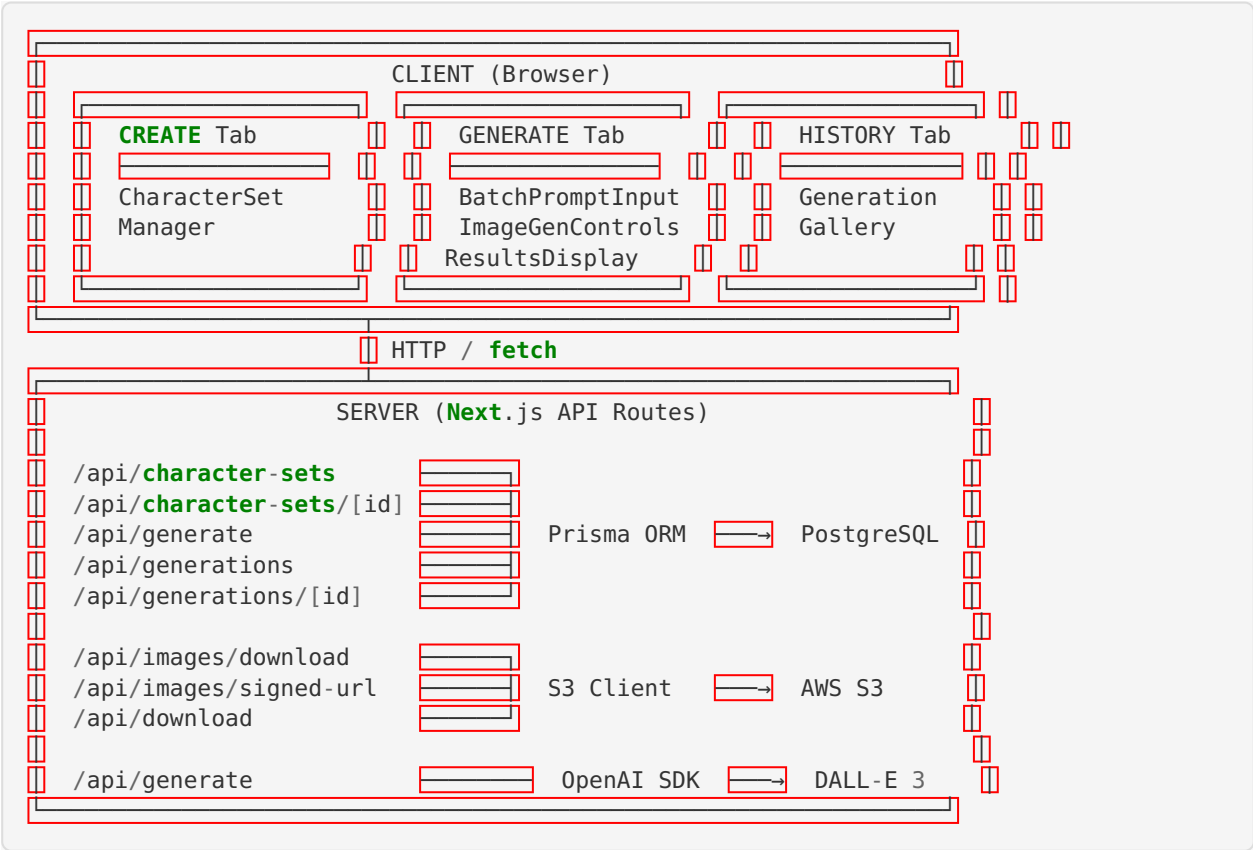
AI-Powered Consistent Character Image Story Creator

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1. SYSTEM ARCHITECTURE



Data Flow

Image Generation Flow

```
User selects character set
↓
Sample prompts auto-populate in textarea
↓
User edits/adds prompts + selects aspect ratio
↓
POST /api/generate { prompts[], characterSetId, aspectRatio }
↓
Server creates Generation record in DB
↓
For each prompt:
  [ ] Fetch reference image URLs from S3 (signed)
  [ ] Enhance prompt with style consistency instructions
  [ ] Call OpenAI DALL-E 3 API
  [ ] Download generated image
  [ ] Upload to S3
  [ ] Create GeneratedImage record in DB
↓
Return generation ID + image metadata
↓
Client displays results in grid
```

Download Flow

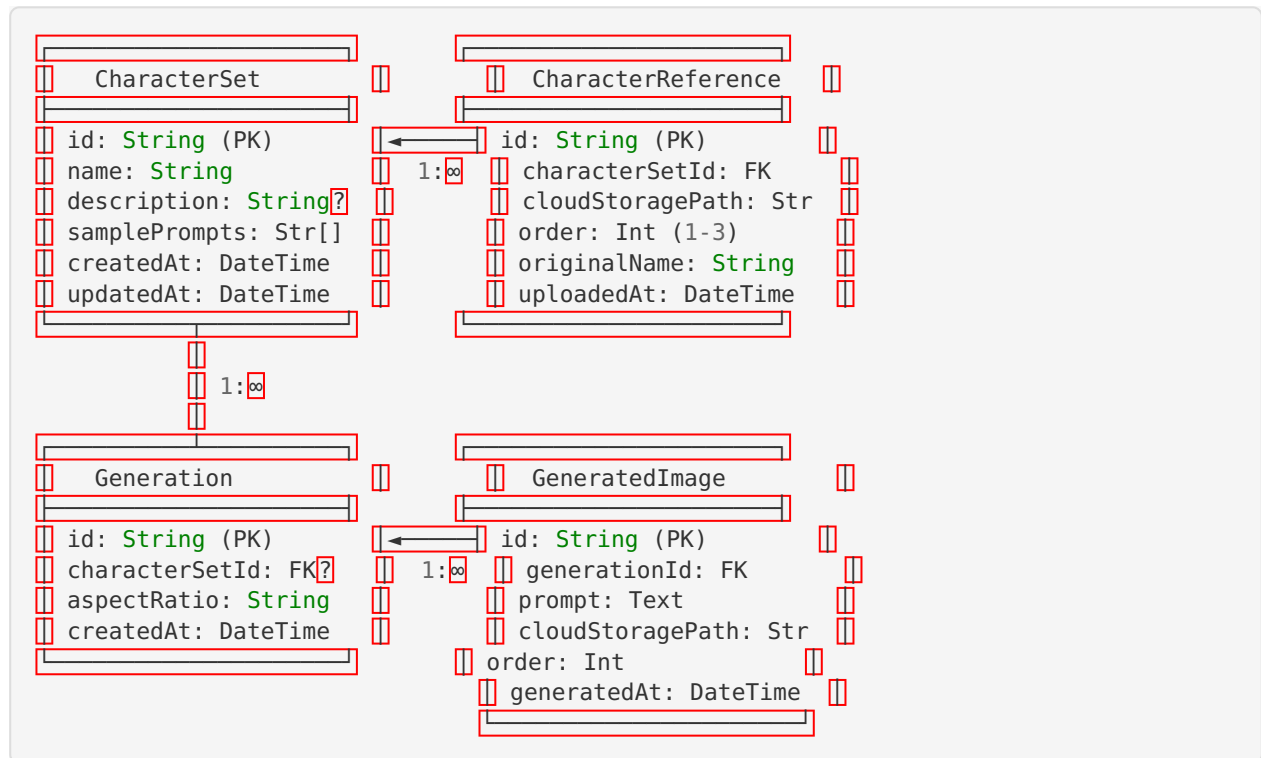
```
User clicks Download button on image
↓
GET /api/images/download?imageId=xxx&filename=001_13Apr2026.png
↓
Server looks up image in DB
↓
Server generates signed S3 URL
↓
Server returns 302 redirect to signed URL
↓
Browser downloads file
```

2. TECHNOLOGY STACK

Category	Technology	Version	Purpose
Framework	Next.js	14.2.28	Full-stack React framework
Language	TypeScript	Latest	Type safety
Runtime	Node.js	18+	Server runtime
UI Framework	React	18.x	Component library
CSS	Tailwind CSS	3.x	Utility-first styling
Components	shadcn/ui	Latest	Pre-built accessible components
Icons	Lucide React	Latest	SVG icon library
ORM	Prisma	6.7.0	Type-safe database access
Database	PostgreSQL	Hosted	Persistent data storage
Cloud Storage	AWS S3	SDK v3	Image file storage
AI/ML	OpenAI DALL-E 3	API v1	Image generation
Package Manager	Yarn	1.x	Dependency management
Theming	next-themes	Latest	Dark/light mode
Archiving	archiver	Latest	ZIP file creation

3. DATABASE SCHEMA

Entity Relationship Diagram



Full Prisma Schema

```

generator client {
  provider = "prisma-client-js"
  binaryTargets = ["native", "linux-musl-arm64-openssl-3.0.x"]
}

datasource db {
  provider = "postgresql"
  url      = env("DATABASE_URL")
}

model CharacterSet {
  id          String          @id @default(cuid())
  name        String
  description  String?
  samplePrompts String[]      @default([])
  createdAt   DateTime        @default(now())
  updatedAt   DateTime        @updatedAt
  references   CharacterReference[]
  generations  Generation[]
  @@index([createdAt])
}

model CharacterReference {
  id              String          @id @default(cuid())
  characterSetId  String
  cloudStoragePath String
  order          Int             // 1, 2, or 3
  originalName   String
  uploadedAt     DateTime        @default(now())
  characterSet    CharacterSet @relation(fields: [characterSetId], references: [id],
onDelete: Cascade)
  @@index([characterSetId])
}

model Generation {
  id          String          @id @default(cuid())
  characterSetId String?
  aspectRatio String          // "16:9", "1:1", "9:16", "4:3"
  createdAt   DateTime        @default(now())
  characterSet CharacterSet? @relation(fields: [characterSetId], references: [id],
onDelete: SetNull)
  images      GeneratedImage[]
  @@index([createdAt])
  @@index([characterSetId])
}

model GeneratedImage {
  id          String          @id @default(cuid())
  generationId String
  prompt      String          @db.Text
  cloudStoragePath String
  order       Int
  generatedAt  DateTime        @default(now())
  generation   Generation @relation(fields: [generationId], references: [id], onDelete: Cascade)
  @@index([generationId])
  @@index([generatedAt])
}

```

Current Database Stats

- CharacterSet: 5 rows
- CharacterReference: 12 rows
- GeneratedImage: 23 rows
- Generation: 6 rows
- **Total: 46 rows across 4 tables**

4. API REFERENCE

GET /api/character-sets

Returns all character sets with their references and sample prompts.

Response:

```
[
  {
    "id": "cuid_xxx",
    "name": "Modern Detective",
    "description": "A sharp-minded investigator solving mysteries",
    "samplePrompts": [
      "Investigating a crime scene in the rain",
      "Examining clues under a desk lamp",
      "Intense chase scene through city streets",
      "Reviewing evidence on a bulletin board",
      "Standing in shadows of a dark alley"
    ],
    "createdAt": "2025-11-26T...",
    "updatedAt": "2025-11-26T...",
    "references": [
      {
        "id": "cuid_yyy",
        "cloudStoragePath": "references/xxx/detective_01.jpg",
        "order": 1,
        "originalName": "detective_01.jpg",
        "uploadedAt": "2025-11-26T..."
      }
    ]
  }
]
```

POST /api/character-sets

Creates a new character set with reference images.

Request: multipart/form-data

- name (string, required)
- description (string, optional)
- images (files, up to 3)

Response: 201 with created character set

GET /api/character-sets/[id]

Returns a single character set with references.

DELETE /api/character-sets/[id]

Deletes a character set and all associated references (cascade).

POST /api/generate

Generates images from prompts using DALL-E 3.

Request:

```
{
  "prompts": ["Scene 1 description", "Scene 2 description"],
  "characterSetId": "cuid_xxx",
  "aspectRatio": "16:9"
}
```

Response:

```
{
  "generationId": "cuid_zzz",
  "images": [
    {
      "id": "cuid_img",
      "prompt": "Scene 1 description",
      "cloudStoragePath": "generations/xxx/001.png",
      "url": "https://i0.wp.com/shakespeareillustration.org/wp-content/uploads/2016/08/hcseulousmov3.jpg?resize=1200%2C1200&ssl=1",
      "order": 1
    }
  ]
}
```

GET /api/generations

Returns all generations with images and character set info.

GET /api/generations/[id]

Returns a single generation with all images.

GET /api/images/download

Server-side download proxy to bypass CORS.

Query Params:

- `imageId` (string, required)
- `filename` (string, optional)

Response: 302 redirect to signed S3 URL

GET /api/images/signed-url

Generates a temporary signed URL for an image.

Query Params:

- `path` (string, required) — S3 cloud storage path

Response:

```
{ "url": "https://s3.../signed-url" }
```

GET /api/download

Downloads a generation as a ZIP file.

Query Params:

- `generationId` (string, required)
- `prefix` (string, optional, default: "story")

Response: ZIP file stream

5. COMPONENT ARCHITECTURE

Component Tree

```

RootLayout (app/layout.tsx)
├── ThemeProvider
├── Page (app/page.tsx)
│   ├── Header (Citability logo + title)
│   ├── Hero Section
│   └── ImageCreator (components/image-creator.tsx) [CLIENT]
│       ├── Tab Navigation (Create | Generate | History)
│       ├── [CREATE] CharacterSetManager
│       │   ├── Character Set Cards (grid)
│       │   ├── New Set Form (dialog)
│       │   └── Skeleton / EmptyState
│       ├── [GENERATE]
│       │   ├── CharacterSetManager (compact selector)
│       │   ├── BatchPromptInput (textarea)
│       │   ├── ImageGenerationControls (aspect ratio + generate button)
│       │   └── GenerationResultsDisplay (grid + preview + download)
│       └── [HISTORY] GenerationGallery
│           ├── Generation Cards
│           ├── Image Grid with Hover Overlay
│           └── Skeleton / EmptyState

```

Key Components

Component	File	Purpose	State
ImageCreator	<code>components/image-creator.tsx</code>	Main controller, tab management	selectedCharacterSet, prompts, aspectRatio, activeTab
CharacterSetManager	<code>components/character-set-manager.tsx</code>	CRUD for character sets	characterSets[], loading, creating
BatchPromptInput	<code>components/batch-prompt-input.tsx</code>	Multi-line prompt editor	inputText (synced with props)
ImageGenerationControls	<code>components/image-generation-controls.tsx</code>	Settings + generate button	loading, progress
GenerationResultsDisplay	<code>components/generation-results-display.tsx</code>	Results grid + downloads	downloading states
GenerationGallery	<code>components/generation-gallery.tsx</code>	History browser	generations[], loading

6. IMAGE GENERATION PIPELINE

`lib/image-generation.ts`

Aspect Ratio Mapping:

```
16:9 → 1792x1024
9:16 → 1024x1792
1:1 → 1024x1024
4:3 → 1024x1024 (closest supported)
```

API Key Source: `/home/ubuntu/.config/abacusai_auth_secrets.json`

```
{
  "openai": {
    "secrets": {
      "api_key": {
        "value": "sk-proj-..."
      }
    }
  }
}
```

Prompt Enhancement:

When reference images are provided, the prompt is enhanced with style consistency instructions to maintain character appearance across generations.

API Call:

```
POST https://api.openai.com/v1/images/generations
{
  model: "dall-e-3",
  prompt: "<enhanced prompt>",
  n: 1,
  size: "1792x1024",
  quality: "standard"
}
```

7. FILE STORAGE (S3)

lib/aws-config.ts

Configures S3 client with:

- AWS_BUCKET_NAME
- AWS_REGION
- AWS_PROFILE
- AWS_FOLDER_PREFIX

lib/s3.ts

Function	Purpose
uploadFile(buffer, key)	Upload image buffer to S3
getSignedDownloadUrl(key)	Generate temporary download URL
deleteFile(key)	Remove file from S3
downloadFile(key)	Retrieve file from S3

Storage Structure

```
<prefix>/
  references/
    <characterSetId>/
      hero_01.jpg
      hero_02.jpg
      hero_03.jpg
  generations/
    <generationId>/
      001.png
      002.png
      003.png
```

8. ENVIRONMENT CONFIGURATION

.env File

```

DATABASE_URL=postgresql://... # PostgreSQL connection
AWS_PROFILE=...               # AWS credential profile
AWS_REGION=...                 # S3 bucket region
AWS_BUCKET_NAME=...           # S3 bucket name
AWS_FOLDER_PREFIX=...         # S3 folder prefix
ABACUSAI_API_KEY=...          # Abacus AI API key

```

Secrets Config

```

~/.config/abacusai_auth_secrets.json
└─ openai.secrets.api_key.value = "sk-proj-..."

```

9. BUILD & DEPLOYMENT

Development

```

cd /home/ubuntu/robert_tech_image_creator/nextjs_space
yarn dev                # Start dev server on port 3000

```

Production Build

```

yarn run build          # Creates optimized production build

```

Build Output (Last Build)

Route (app)	Size	First Load JS
f /	39.4 kB	138 kB
f /_not-found	872 B	88.1 kB
f /api/character-sets	0 B	0 B
f /api/character-sets/[id]	0 B	0 B
f /api/download	0 B	0 B
f /api/generate	0 B	0 B
f /api/generations	0 B	0 B
f /api/generations/[id]	0 B	0 B
f /api/images/download	0 B	0 B
f /api/images/signed-url	0 B	0 B
+ First Load JS shared by all	87.2 kB	

Deployment

- Currently: **NOT DEPLOYED** to any public URL
- Deploy via Abacus AI Deploy button
- Gets `.abacusai.app` subdomain automatically

10. FILE STRUCTURE

```

robert_tech_image_creator/
├── BUILD_SHEET.md
├── MASTER_BUILD_SHEET_CITABILITY.md
├── CONVERSATION_TRANSCRIPT_CITABILITY.md
├── TECHNICAL_DOCUMENTATION_CITABILITY.md
├── TASK_CHECKLIST_CITABILITY.md
├── QUICK_REFERENCE_CITABILITY.md
├── nextjs_space/
│   ├── .env
│   ├── next.config.js
│   ├── tailwind.config.ts
│   ├── tsconfig.json
│   ├── postcss.config.js
│   └── app/
│       ├── globals.css
│       ├── layout.tsx
│       ├── page.tsx
│       └── api/
│           ├── character-sets/
│           │   ├── route.ts
│           │   ├── [id]/route.ts
│           │   ├── generate/route.ts
│           │   ├── generations/
│           │   │   ├── route.ts
│           │   │   ├── [id]/route.ts
│           │   └── images/
│           │       ├── download/route.ts
│           │       ├── signed-url/route.ts
│           │       └── download/route.ts
│           ├── components/
│           │   ├── image-creator.tsx
│           │   ├── character-set-manager.tsx
│           │   ├── batch-prompt-input.tsx
│           │   ├── image-generation-controls.tsx
│           │   ├── generation-results-display.tsx
│           │   ├── generation-gallery.tsx
│           │   ├── theme-provider.tsx
│           │   └── ui/ (30+ shadcn components)
│           └── lib/
│               ├── aws-config.ts
│               ├── s3.ts
│               ├── db.ts
│               ├── types.ts
│               ├── image-generation.ts
│               └── utils.ts
│       ├── hooks/use-toast.ts
│       ├── prisma/schema.prisma
│       ├── scripts/
│       │   ├── seed.ts
│       │   ├── add-reporter.ts
│       │   └── safe-seed.ts
│       └── public/
│           ├── favicon.svg
│           ├── og-image.png
│           └── test-images/ (12 reference images)

```

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